Review of Last Lecture

6.5 Zero-State Response and Zero-Input Response of Discrete-Time Systems

6.6 Unit Sample Response of Discrete-Time Systems

6.7 Convolution Sum

1. Zero-Input Response

- **Definition of Zero-Input Response**: The response generated solely by the initial state of the system (with the input signal being zero), denoted as $\boldsymbol{y_{zi}(n)}$.

- Mathematical Model:

$$\sum_{k=0}^{N} a_k y(n-k) = 0$$

- Determining coefficients of the homogeneous solution for **zero-input response**: The n in the initial condition $\boldsymbol{y(n)}$ can be either **positive or negative** and is **not necessarily continuous**. **$y(-1)$, $y(-2)$,…$y(-N)$ or $y(0)$, $y(1)$,…$y(N-1)$** can all be used as **boundary conditions.** For $\boldsymbol{y(n)}$ when $\boldsymbol{n\geq 0}$**,** it can be obtained by iteratively substituting $\boldsymbol{y(n)}$ ($n<0$) into the difference equation, and **the right-hand side of the difference equation must be set to zero during iteration**.

2. Zero-State Response

➢**Definition of Zero-State Response**: The zero-state response generated by the excitation $\boldsymbol{x(n)}$ when the initial state of the system is zero, denoted as $\boldsymbol{y_{zs}(n)}$.

➢Mathematical Model：$$\sum_{k=0}^{N} a_k y(n-k) = \sum_{r=0}^{M} b_r x(n-r)$$

➢Solution :
$$y_{zs}(n) = \sum_{k=1}^{N} C_{zsk} \alpha_k^n + D(n)$$

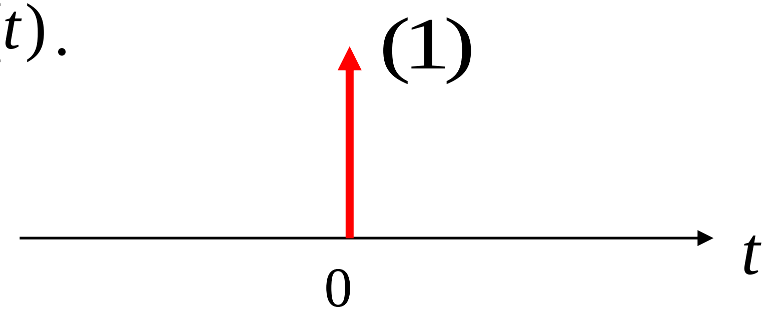
➢ Determining coefficients of the homogeneous solution for **zero-state response**: **Let $y(-1)=y(-2)=\ldots=y(-N+1)=0$,** and iteratively find $\boldsymbol{y(0), y(1), \ldots}$. **Assuming the excitation is applied at $n = 0$, the boundary conditions must include at least one $y(n)$ when $n \geq 0$, and n must be continuous when $n < 0$** (to ensure $\boldsymbol{y(n)}$ with $\boldsymbol{n \geq 0}$ can be obtained by iteration) . For example, $\boldsymbol{y(0)}$, $\boldsymbol{y(-1)}$, $\boldsymbol{\ldots y(-N+1)}$ **can be used as boundary conditions**.

6.6 Unit Sample Response of Discrete-Time Systems

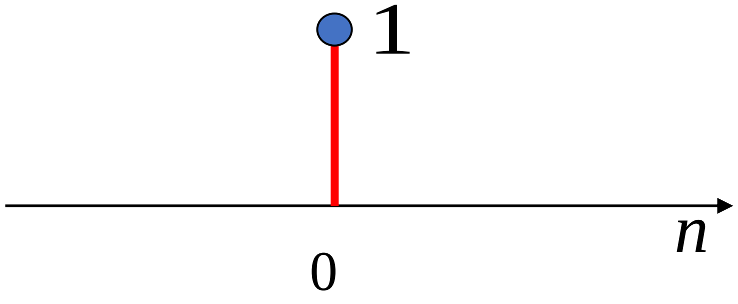
Unit Sample Response: It is the **zero-state response** of the system when the excitation is $\delta(n)$, denoted by $\boldsymbol{h(n)}$.

Note the difference between $\delta(n)$ and $\delta(t)$.

$$\begin{cases} \int_{-\infty}^{\infty} \delta(t)dt = 1 & (t = 0) \\ \delta(t) = 0 & (t \neq 0) \end{cases}$$



$$\delta(n) = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$$



Solution of Unit Sample Response:

Convert $\delta(n)$ into boundary conditions, so finding the unit sample response is converted into finding **the homogeneous solution**, i.e., **the zero-input response when n>0.**

Example: Given the difference equation model of a system, find the unit sample response.

$$y(n)-5y(n-1)+6y(n-2)=x(n)\text{-}3x(n\text{-}2)$$

Solution:

(1) The homogeneous solution is $C_1 2^n + C_2 3^n$

(2) Consider only the unit sample response $\boldsymbol{h_1(n)}$ of the system when the excitation $\boldsymbol{x(n)=\delta(n)}$ acts.

$$h_1(0)=1, \quad h_1(-1)=0 \qquad C_1=-2, \quad C_2=3$$

$$h_1(n)=(3^{n+1}-2^{n+1})u(n)$$

(3) Consider only the excitation $\boldsymbol{-3x(n-2)}$

By the LTI property

$$h_2(n)=-3h_1(n-2)=-3(3^{n-1}-2^{n-1})u(n-2)$$

(4)

$$h(n)=h_1(n)+h_2(n)=(3^{n+1}-2^{n+1})u(n)-3(3^{n-1}-2^{n-1})u(n-2)$$

$$=(3^{n+1}-2^{n+1})\left[\delta(n)+\delta(n-1)+u(n-2)\right]-(3^n-3\times 2^{n-1})u(n-2)$$

$$=\delta(n)+5\delta(n-1)+(2\times 3^n-2^{n-1})u(n-2)$$

6.6 Unit Sample Response of Discrete-Time Systems

Analyzing Causality and Stability of Systems Based on Unit Sample Response

Causal System: A system where the output change does not precede the input change. That is, the response depends only on the current and previous excitations.

The **necessary and sufficient condition** for a discrete-time linear time-invariant system to be **a causal system** is:

$$h(n)=0\ (n<0)\ \text{or}\ \ h(n)=h(n)u(n)$$

Stable System: If the input is bounded, the output must be bounded.

The **necessary and sufficient condition** for a discrete practical linear time-invariant system to be **a stable system** is:

$$\sum_{n=-\infty}^{\infty}\left|h(n)\right|\le M\,, M\ \ \text{is a bounded positive vaule}$$

6.7.1 Definition and Physical Meaning of Convolution Sum

The convolution sum of any signals $x_1(n)$ and $x_2(n)$is defined as :

$$x(n) = x_1(n) * x_2(n)= \sum_{m=-\infty}^{\infty} x_1(m)x_2(n-m)$$

6.7.2 Finding Zero-State Response Using Convolution Sum

$$y_{zs}(n) = x(n) * h(n) = \sum_{m=-\infty}^{\infty} h(m)x(n-m)$$

The zero-state response of a discrete-time LTI system is equal to the convolution sum of the excitation signal and the unit sample response.

6.7.3 Operation Method of Convolution Sum

$$y(n) = x(n) * h(n) = \sum_{m=-\infty}^{\infty} x(m)h(n-m)$$

The graphical method of convolution sum consists of five steps:

(1) Variable Substitution: Change the independent variable in ***x*(*n*)** and ***h*(*n*)** from ***n*** to ***m*,** where ***m*** becomes the independent variable of the function.

(2) Reversing: Fold one of the signals (usually the simpler one)

$$h(m) \xrightarrow{\text{Reverse}} h(-m)$$

(3) Shifting: Shift the folded signal

$$h(-m) \xrightarrow{\text{shift by } n} h(-(m-n)) = h(n-m)$$

(4) Multiplying: Multiply ***x*(*m*)** by ***h*(*n-m*)**

(5) Summing: Sum the product graph.

6.7 Convolution Sum

Given a discrete-time LTI system, its unit sample response $\boldsymbol{h(n)}$ has non-zero values only in the interval $N_0 \le n \le N_1$, and the input $\boldsymbol{x(n)}$ has non-zero values only in the interval of $N_2 \le n \le N_3$. Then, the zero-state response has non-zero values only in the interval of $N_4 \le n \le N_5$, where

$$\boxed{N_4 = N_0 + N_2 \quad N_5 = N_1 + N_3}$$

Assume the unit sample response $\boldsymbol{h(n)}$ has non-zero values only in $N_0 \le n \le N_1$, with sequence length $L_1 = N_1 - N_0 + 1$. The input $\boldsymbol{x(n)}$ has non-zero values only in $N_2 \le n \le N_3$, with sequence length $L_2 = N_3 - N_2 + 1$.

The zero-state response $\boldsymbol{y(n)}$ has non-zero values only in $N_0 + N_2 \le n \le N_1 + N_3$, with **sequence length L_3**:

$$\boxed{L_3 = L_1 + L_2 - 1}$$

For continuous-time signals, the length of the convolution integral is $L_3 = L_1 + L_2$.

单选题 1分

设置

Calculate the convolution sum of $x(n)=\{2,1,\overset{\downarrow}{3},0,\ 4\}$ and $h(n)=\{2,\overset{\downarrow}{3},1,\ 4\}$

A $y(n)=\{4,8,\overset{\downarrow}{11},18,15,\ 24,\ 4,16\}$

B $y(n)=\{4,\ 7,\overset{\downarrow}{11},18,15,\ 24,\ 4,16\}$

C $y(n)=\{4,8,12,\ \overset{\downarrow}{17},15,\ 24,\ 4,16\}$

D $y(n)=\{4,8,11,\overset{\downarrow}{18},16,\ 23,\ 4,16\}$

提交

6.7.4 Properties of Convolution Sum

Commutative Law

$$x_1(n)*x_2(n)=x_2(n)*x_1(n)$$

Distributive Law

$$x_1(n)*\left[x_2(n)+x_3(n)\right]=x_1(n)*x_2(n)+x_1(n)*x_3(n)$$

Associative Law

$$x_1(n)*\left[x_2(n)*x_3(n)\right]=\left[x_1(n)*x_2(n)\right]*x_3(n)$$

Convolution with Unit Sample Sequence

$$x(n)*\delta(n)=x(n)$$

$$x(n)*\delta(n-n_0)=x(n-n_0)$$

Convolution with Step Sequence

$$x(n)*u(n)=\sum_{m=-\infty}^{n}x(m)$$

6.7.5 Finding Zero-State Response of a System Using Convolution Sum

Example 6-10: The difference equation of a discrete-time LTI system is

$$y(n)+3y(n-1)+2y(n-2)=x(n)$$

Given the excitation $x(n)=3\left(\frac{1}{2}\right)^n u(n)$, find the zero-state response $\boldsymbol{y_{zs}(n)}$.

Solution:

Step 1: Find the unit sample response of the system

$$h(n)+3h(n-1)+2h(n-2)=\delta(n)$$

The characteristic roots are **-1 & -2**, $\therefore h(n)=\left[C_1(-1)^n+C_2(-2)^n\right]u(n)$

$h(-1)=0,\ h(-2)=0,$ iterate to find $h(0)=1.$

Using $h(0)$ and $h(-1)$, get $C_1=-1,\ \ C_2=2.$

$$\therefore h(n)=\left[-(-1)^n+2(-2)^n\right]u(n)$$

Step 2: Find the convolution sum of the excitation signal and the unit sample response to obtain the zero-state response.

$$y_{zs}(n)=\sum_{m=-\infty}^{\infty}x(m)h(n-m)$$

$$=\sum_{m=-\infty}^{\infty}3(\frac{1}{2})^{m}u(m)\cdot[-(-1)^{n-m}+2(-2)^{n-m}]u(n-m)$$

$$=\begin{cases}-3(-1)^{n}\sum_{m=0}^{n}(-\frac{1}{2})^{m}+6(-2)^{n}\sum_{m=0}^{n}(-\frac{1}{4})^{m}, & n\geq 0\\ 0 & n<0\end{cases}$$

$$=[-2(-1)^{n}+\frac{24}{5}(-2)^{n}+\frac{1}{5}(\frac{1}{2})^{n}]u(n)$$

Contents of This Lecture

Learning Objectives of This Lecture

1. Understand the relationship between **z-transform** and **Laplace transform;**
2. Be familiar with the z-transforms and their Regions of Convergence (ROC) of **common sequences**;
3. Be able to analyze the **ROC** of various types of sequences;
4. Skillfully master the **method of partial fractions** to find the **inverse z-transform**

The z-transform is introduced with the help of the Laplace transform of sampled signals.

For a continuous causal signal ***x*(*t*)** subjected to uniform impulse sampling, the expression of the sampled signal ***x*$_s$(*t*)** is

$$x_s(t) = x(t)\delta_T(t) = \sum_{n=0}^{\infty} x(nT)\delta(t - nT)$$

where ***T*** is the sampling interval.

Unilateral Laplace Transform

$$X_s(s) = \int_0^{\infty} x_s(t)e^{-st}dt = \sum_{n=0}^{\infty} x(nT)\int_0^{\infty} \delta(t - nT)e^{-st}dt = \sum_{n=0}^{\infty} x(nT)e^{-snT}$$

Let $z = e^{sT}$,where **z is a complex variable**. The relationship between the Laplace transform of the sampled signal and the z-transform of the sequence is as follows:

$$\boxed{X(z)\Big|_{z=e^{sT}} = X_s(s)}$$

7.1 Definition of z-Transform

$$z = e^{sT}\ ,\quad X(z) = \sum_{n=0}^{\infty} x(nT) z^{-n}$$

Usually let T=1, $z = e^{s}$

$$\boxed{X(z) = \sum_{n=0}^{\infty} x(n) z^{-n}}$$

Unilateral z-Transform

$$\boxed{X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}}$$

Bilateral z-Transform

The z-transform of a sequence is **a power series of the complex variable z^{-1}**, and **its coefficients are $x(n)$.**

The unilateral and bilateral z-transforms of a causal sequence are equivalent.

We focus on the unilateral, while also considering the bilateral.

7.1 Definition of Z-Transform

Why is **z-transform** introduced?

- The z-transform is a powerful tool for **solving difference equations** (converting difference equations into algebraic equations), similar to the Laplace transform in continuous-time systems.
- The z-transform is also used in **the analysis and design of digital filters**, as well as various types of digital signal processing problems.

Generation and Development of z-Transform:

- Preliminary understanding in the 18th century;
- Further development and applications in the 1950s–1960s.

单选题 1分

设置

The z-transform of a sequence is meaningful under any conditions.

The statement is ( )

A True

B False

提交

Unilateral (Bilateral) z-Transform

1. z-Transform of the Unit Sample Sequence

$$ZT[\delta(n)] = \sum_{n=0}^{\infty} \delta(n) z^{-n} = 1 \quad (0 \le |z| \le \infty, \ \text{Convergent on the entire plane})$$

$$ZT[\delta(n-1)] = \sum_{n=0}^{\infty} \delta(n-1) z^{-n} = z^{-1} \quad (|z| > 0)$$

Unilateral (Bilateral) z-Transform

$$ZT[\delta(n+1)] = \sum_{n=-\infty}^{-1} \delta(n+1) z^{-n} + \sum_{n=0}^{\infty} \delta(n+1) z^{-n} = z^{1} + 0 = z \quad (|z| < \infty)$$

Bilateral z-Transform

2. z-Transform of the Unit Step Sequence

$$ZT[u(n)] = \sum_{n=0}^{\infty} u(n) z^{-n} = \sum_{n=0}^{\infty} z^{-n} = \boxed{\frac{1}{1-z^{-1}} = \frac{z}{z-1} \qquad (|z^{-1}| < 1 \rightarrow |z| > 1)}$$

Unilateral (Bilateral) z-Transform

单选题 1分

Setting

The z-transform of the sequence $a^n u(n)$ $(|a| \neq 1)$ is ( )

A $\frac{z}{a-z}$ $(|z| < |a|)$

B $\frac{z}{a-z}$ $(|z| > |a|)$

C $\frac{z}{z-a}$ $(|z| < |a|)$

D $\frac{z}{z-a}$ $(|z| > |a|)$

提交

3. z-Transform of the Unilateral Exponential Sequence

$$ZT[a^n u(n)] = \sum_{n=0}^{\infty} a^n z^{-n} = \sum_{n=0}^{\infty} (az^{-1})^n = \boxed{\frac{1}{1-az^{-1}} = \frac{z}{z-a} \quad (|az^{-1}| < 1 \rightarrow |z| > |a|)}$$

4. z-Transform of the Ramp Sequence

$$ZT[nu(n)] = \sum_{n=0}^{\infty} nu(n) z^{-n} = \sum_{n=0}^{\infty} nz^{-n}$$

From $\displaystyle\sum_{n=0}^{\infty} (z^{-1})^n = \frac{1}{1-z^{-1}} \quad (|z| > 1)$ （1）

Differentiate both sides of (1) with respect to $\mathbf{z^{-1}}$, $\displaystyle\sum_{n=0}^{\infty} n(z^{-1})^{n-1} = \frac{1}{(1-z^{-1})^2}$ （2）

Multiply both sides of (2) by $\mathbf{z^{-1}}$, $\displaystyle ZT[nu(n)] = \sum_{n=0}^{\infty} nz^{-n} = \frac{z}{(z-1)^2} \quad (|z| > 1)$

5. z-Transform of the Unilateral Complex Exponential Sequence

$$ZT[e^{j\omega_0 n}u(n)] = \frac{z}{z-e^{j\omega_0}} \qquad |z| > \left|e^{j\omega_0}\right| = 1$$

$$ZT[e^{-j\omega_0 n}u(n)] = \frac{z}{z-e^{-j\omega_0}} \qquad |z| > \left|e^{-j\omega_0}\right| = 1$$

$$ZT[\beta^n e^{j\omega_0 n}u(n)] = \frac{z}{z-\beta e^{j\omega_0}} \qquad |z| > \left|\beta e^{j\omega_0}\right| = |\beta|$$

$$ZT[\beta^n e^{-j\omega_0 n}u(n)] = \frac{z}{z-\beta e^{-j\omega_0}} \qquad |z| > \left|\beta e^{-j\omega_0}\right| = |\beta|$$

6. z-Transform of the Unilateral Sinusoidal Sequence

$$ZT[\cos(\omega_0 n)u(n)] = ZT[(e^{j\omega_0 n} + e^{-j\omega_0 n})/2 \cdot u(n)]$$

$$= \left(\frac{z}{z - e^{j\omega_0}} + \frac{z}{z - e^{-j\omega_0}}\right)/2$$

$$\boxed{= \frac{z(z - \cos\omega_0)}{z^2 - 2z\cos\omega_0 + 1} \qquad (|z| > 1)}$$

$$ZT[\sin(\omega_0 n)u(n)] = ZT[(e^{j\omega_0 n} - e^{-j\omega_0 n})/(2j) \cdot u(n)]$$

$$= \left(\frac{z}{z - e^{j\omega_0}} - \frac{z}{z - e^{-j\omega_0}}\right)/2j$$

$$\boxed{= \frac{z\sin\omega_0}{z^2 - 2z\cos\omega_0 + 1} \qquad (|z| > 1)}$$

7. z-Transform of the Unilateral Exponential Sinusoidal Sequence

$$ZT[\beta^n \cos(\omega_0 n)u(n)] = ZT[\beta^n (e^{j\omega_0 n} + e^{-j\omega_0 n})/2]$$

$$= (\frac{z}{z-\beta e^{j\omega_0}} + \frac{z}{z-\beta e^{-j\omega_0}})/2$$

$$= \frac{z(z-\beta\cos\omega_0)}{z^2 - 2z\beta\cos\omega_0 + \beta^2} \quad (|z| > |\beta|)$$

$$ZT[\beta^n \sin(\omega_0 n)u(n)] = ZT[\beta^n (e^{j\omega_0 n} - e^{-j\omega_0 n})/2j]$$

$$= (\frac{z}{z-\beta e^{j\omega_0}} - \frac{z}{z-\beta e^{-j\omega_0}})/2j$$

$$= \frac{z\beta\sin\omega_0}{z^2 - 2z\beta\cos\omega_0 + \beta^2} \quad (|z| > |\beta|)$$

7.3 Region of Convergence of z-Transform

$$X(z) = ZT[x(n)] = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

If $x(n)$ is bounded, **the set of all z-values for which the above series converges** is called the **Region of Convergence (ROC)**.

The **necessary and sufficient condition** for the series to converge is satisfying **the absolute summability condition**:

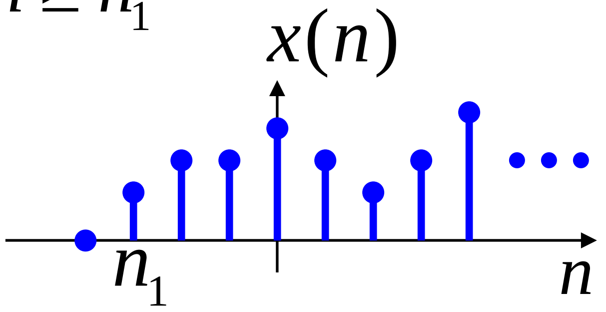
$$\sum_{n=-\infty}^{\infty} \left| x(n) z^{-n} \right| < \infty$$

The left side forms a positive-term series, and common determination methods for the convergence of positive-term series can be used:

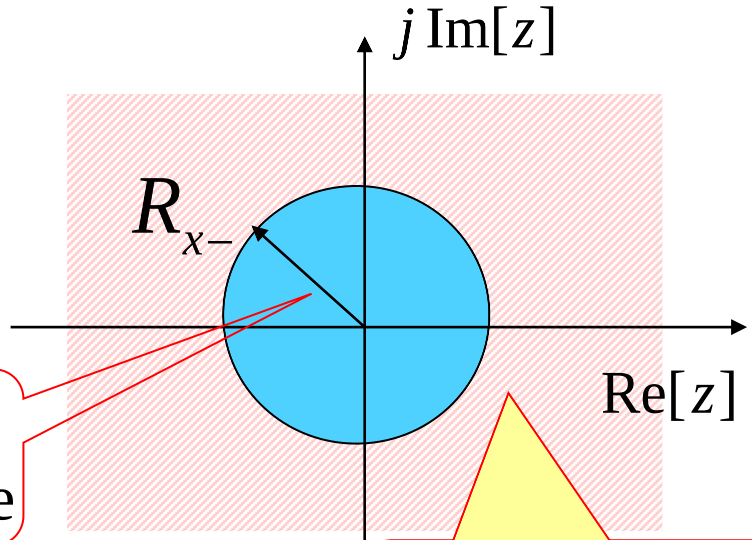
Ratio Test: $\lim_{n \to \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$

Root Test: $\lim_{n \to \infty} \sqrt[n]{|a_n|} < 1$

1. Right-Sided Sequence: A sequence that has non-zero finite values only in the interval $n \geq n_1$

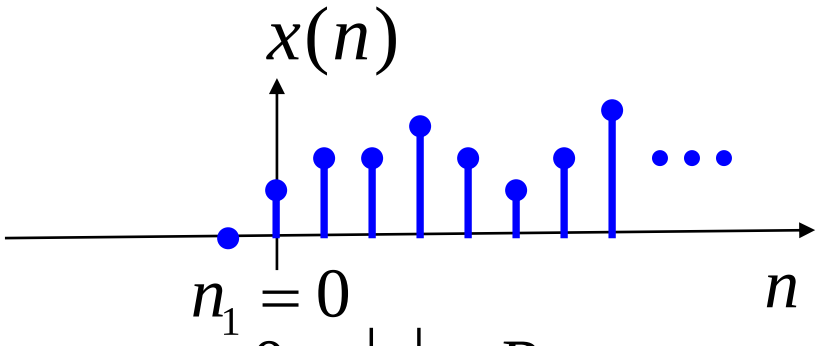


$$X(z) = \sum_{n=n_1}^{\infty} x(n) z^{-n} \qquad n_1 \leq n \leq \infty$$



$n_1 < 0, \quad R_{x-} < |z| < \infty$ (The ROC does not include $|z| = \infty$, because ∞^{-n_1} is infinite)

Radius of Convergence

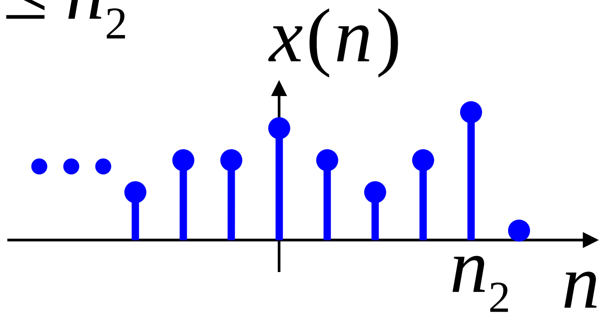


$n_1 = 0, \quad |z| > R_{x-}$

The ROC is outside the circle. If $\boldsymbol{n_1 < 0}$, $|z| = \infty$ is not included.

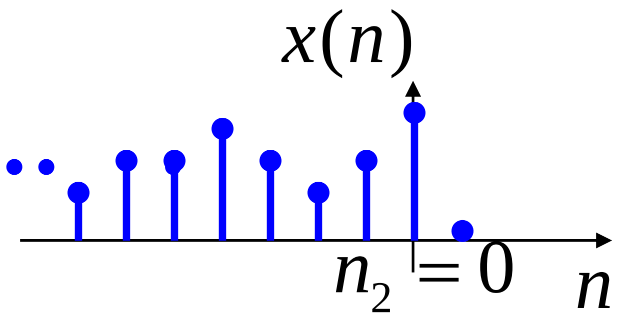
A **causal sequence** is a special case of a right-sided sequence, and its ROC is $|z| > R_{x-}$

2. Left-Sided Sequence: A sequence $x(n)$ that has non-zero finite values only in the interval $n \le n_2$

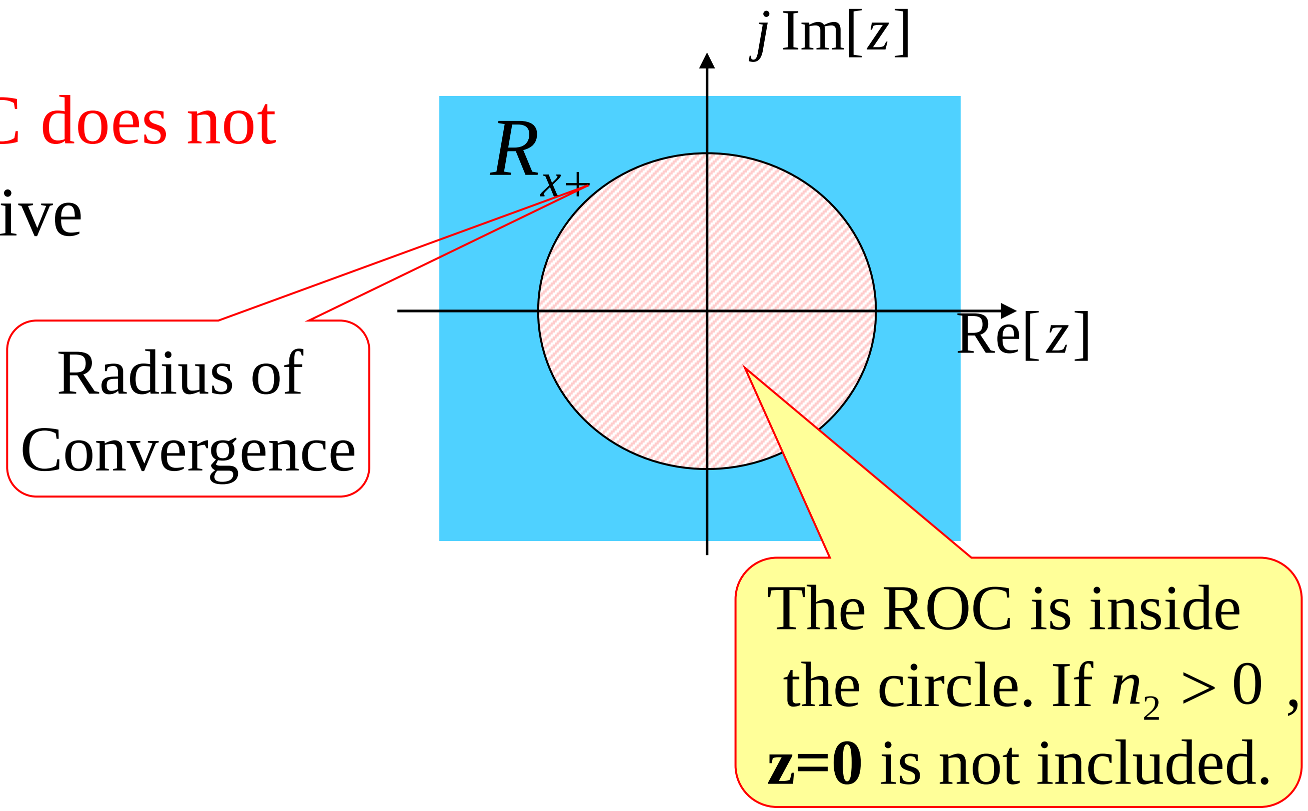


$$X(z) = \sum_{n=-\infty}^{n_2} x(n) z^{-n} \qquad -\infty \le n \le n_2$$

$n_2 > 0, \quad 0 < |z| < R_{x+}$ （The ROC does not include $z = 0$, because the negative power of 0 is meaningless）

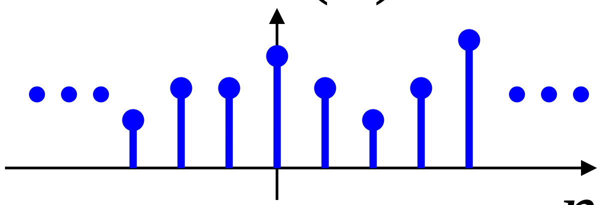


$n_2 = 0, \quad |z| < R_{x+}$



7.3 Region of Convergence of z-Transform

3. Double-Sided Sequence: A sequence that has non-zero finite values in the interval $-\infty \le n \le \infty$

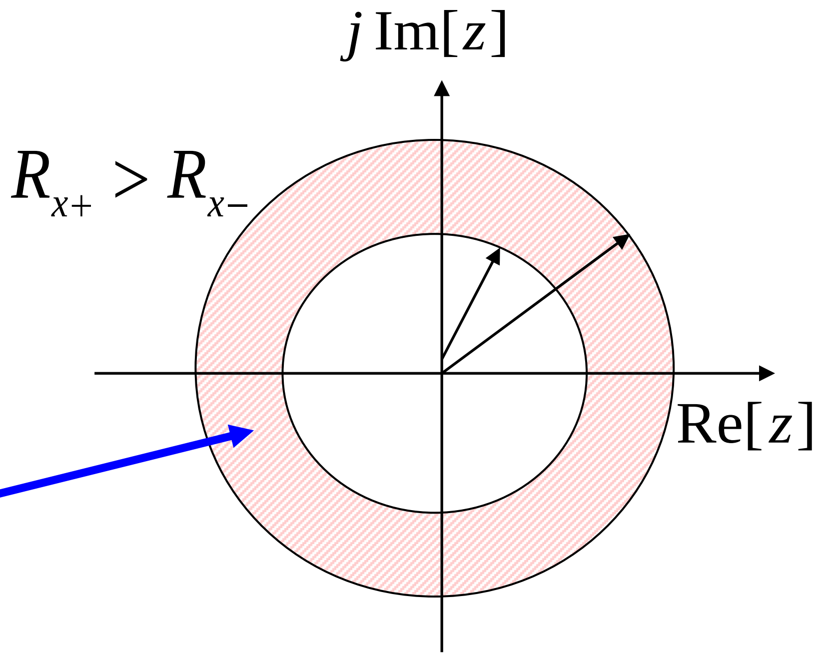


$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n} \qquad -\infty \le n \le \infty$$

$$X(z) = \sum_{n=-\infty}^{-1} x(n) z^{-n} + \sum_{n=0}^{\infty} x(n) z^{-n}$$

Converges inside the circle $|z| < R_{x+}$

Converges outside the circle $|z| > R_{x-}$

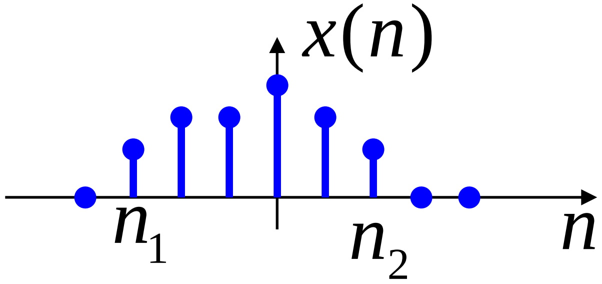


$R_{x+} > R_{x-}$ There is an annular ROC

$R_{x+} < R_{x-}$ No ROC

7.3 Region of Convergence of z-Transform

4. Finite-Length Sequence: A sequence that has non-zero finite values in the finite interval $n_1 \le n \le n_2$



$$X(z) = \sum_{n=n_1}^{n_2} x(n) z^{-n} \qquad n_1 \le n \le n_2$$

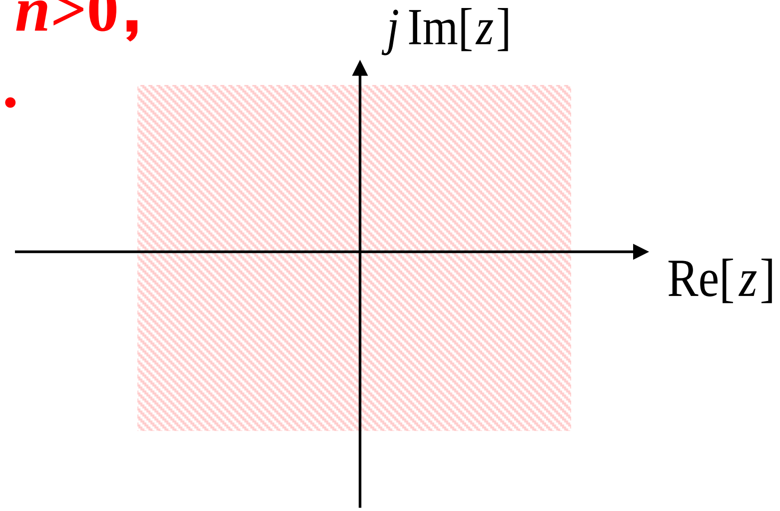
Divided into three cases:

$$n_1 < 0, n_2 > 0 \quad 0 < |z| < \infty$$

$$n_1 \ge 0, n_2 > 0 \quad 0 < |z| \le \infty$$

$$n_1 < 0, n_2 \le 0 \quad 0 \le |z| < \infty$$

0^{-n} is meaningless if n>0,
∞^{-n} is infinite if n<0.



The ROC is at least the entire z-plane but 0 and ∞ (If pole-zero cancellation occurs, the ROC expands to the entire z-plane).

Which of the following statements are correct?（ ）

A The unilateral z-transform’s transform pair has a one-to-one correspondence with the sequence.

B The unilateral z-transform’s has a unique ROC.

C The bilateral z-transform has a one-to-one correspondence with the sequence.

D The bilateral z-transform may correspond to different sequences.

提交

7.3 Region of Convergence of z-Transform

For the unilateral z-transform, there is a one-to-one correspondence between the sequence and the transform expression, and it has a unique ROC.
For the bilateral z-transform, different sequences may map to the same transform expression under different ROCs.

Example: $x_1(n) = a^n u(n)$

$$X_1(z) = \frac{1}{1-az^{-1}} = \frac{z}{z-a} \qquad |z| > |a|$$

$$x_2(n) = -a^n u(-n-1)$$

$$X_2(z) = -\sum_{n=-\infty}^{-1} a^n z^{-n} = 1 - \sum_{n=0}^{\infty} (a^{-1}z)^n$$

$$= 1 - \frac{1}{1-a^{-1}z} = \frac{z}{z-a} \qquad |z| < |a|$$

For the z-transform of a given sequence, **its ROC must be specified**.

Example 8-1: Find the unilateral and bilateral z-transforms of the sequence $x(n)=a^n u(n)-b^n u(-n-1)$ and determine its ROC (where $b>a>0$).

Solution: This is a two-sided sequence.

Unilateral z-transform: $$X(z)=\sum_{n=0}^{\infty}x(n)z^{-n}=\sum_{n=0}^{\infty}a^n z^{-n}=\sum_{n=0}^{\infty}(az^{-1})^n$$

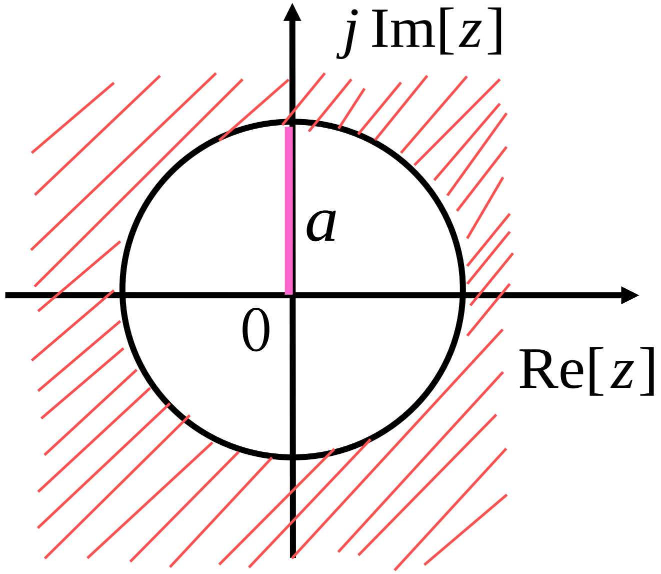
If $|z|>a$, $X(z)=\dfrac{1}{1-az^{-1}}=\dfrac{z}{z-a}$

Bilateral z-transform: $$X(z)=\sum_{n=-\infty}^{\infty}[a^n u(n)-b^n u(-n-1)]z^{-n}$$

$$=\sum_{n=0}^{\infty}a^n z^{-n}-\sum_{n=-\infty}^{-1}b^n z^{-n}=\sum_{n=0}^{\infty}a^n z^{-n}+1-\sum_{n=0}^{\infty}b^{-n}z^n$$

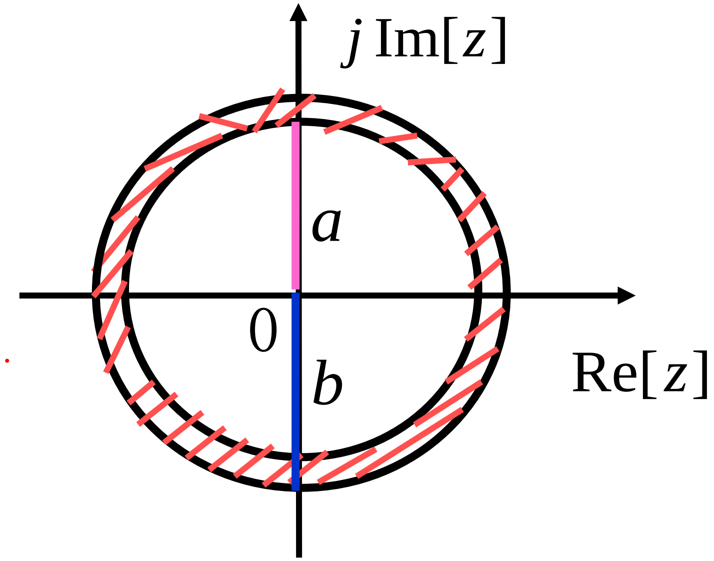
If $a<|z|<b$, $X(z)=\dfrac{1}{1-az^{-1}}+1+\dfrac{b}{1-bz^{-1}}=\dfrac{z}{z-a}+\dfrac{z}{z-b}$

7.3 Region of Convergence of z-Transform



ROC of the sequence's unilateral z-transform

$$X(z)=\frac{z}{z-a}\quad(|z|>a)$$



ROC of the sequence's bilateral z-transform

$$X(z)=\frac{z}{z-a}+\frac{z}{z-b}\quad(a<|z|<b)$$

The ROC typically does not contain any poles (unless zero-pole cancellation occurs), and it is usually bounded by poles. The ROC of a right-sided sequence extends outward from the outermost (largest-magnitude) pole of *X*(z) to z→∞ (may include ∞); The ROC of a left-sided sequence extends inward from the innermost (smallest-magnitude) pole of *X*(z) to z→0 (may include 0).

$$X(z) = ZT[x(n)] = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

Partial Fraction Expansion Method: $X(z) = \dfrac{b_0 + b_1 z + \cdots b_{r-1} z^{r-1} + b_r z^r}{a_0 + a_1 z + \cdots + a_{k-1} z^{k-1} + a_k z^k}$

1. *X(z)* has only first-order poles

$$X(z) = A_0 + \sum_{m=1}^{M} \frac{A_m z}{z - z_m} = A_0 + \frac{A_1 z}{z - z_1} + \cdots + \frac{A_M z}{z - z_M} \qquad A_0 = X(z)\big|_{z=0} = \frac{b_0}{a_0}$$

Let z_m be a first-order pole of $X(z)$ $A_m = \left[(z - z_m) \dfrac{X(z)}{z} \right]_{z=z_m}$ $\quad X(z) = A_0 + \displaystyle\sum_{m=1}^{M} \frac{zA_m}{z - z_m}$

or $\quad X(z) = A_0 + \displaystyle\sum_{m=1}^{M} \frac{A_m}{1 - z_m z^{-1}} = A_0 + \frac{A_1}{1 - z_1 z^{-1}} + \cdots + \frac{A_M}{1 - z_M z^{-1}} \qquad A_m = \left[\left(1 - z_m z^{-1}\right) X(z) \right]_{z=z_m}$

Inverse Transform: $x(n) = A_0 \delta(n) + \displaystyle\sum_{m=1}^{M} A_m z_m^n = A_0 \delta(n) + A_1 z_1^n + \cdots + A_M z_M^n$

单选题 1分

Setting

The inverse transform $x(n)$ of $X(z)=\dfrac{z^2}{z^2-1.5z+0.5}$ (|z|>1) （）

A $(1\text{-}0.5^n)u(n)$

B $(2\text{-}0.5^n)u(n)$

C $2(1\text{-}0.5^n)u(n)$

D $2(1+0.5^n)u(n)$

提交

Example 8-2： Find the inverse transform ***x*(*n*)** of $X(z)=\dfrac{z^2}{z^2-1.5z+0.5}$ **(|z|>1).**

Solution： This is a **right-sided sequence.**

Method 1: $X(z)=\dfrac{z^2}{(z-1)(z-0.5)}$ $\qquad \dfrac{X(z)}{z}=\dfrac{A_1}{z-0.5}+\dfrac{A_2}{z-1}$

where $A_1=\left[\dfrac{X(z)}{z}(z-0.5)\right]_{z=0.5}=-1,\quad A_2=\left[\dfrac{X(z)}{z}(z-1)\right]_{z=1}=2$

then $X(z)=\dfrac{2z}{z-1}-\dfrac{z}{z-0.5}$

$x(n)=(2-0.5^n)u(n)$

Method 2:

$$X(z)=\frac{z^2}{(z-1)(z-0.5)}=\frac{1}{(1-z^{-1})(1-0.5z^{-1})}$$

$$X(z)=\frac{A_1}{1-0.5z^{-1}}+\frac{A_2}{1-z^{-1}}$$

where $A_1=\left[X(z)(1-0.5z^{-1})\right]_{z=0.5}=-1,\quad A_2=\left[X(z)(1-z^{-1})\right]_{z=1}=2$

then $X(z)=\dfrac{2}{1-z^{-1}}-\dfrac{1}{1-0.5z^{-1}}$

$$x(n)=(2\text{-}0.5^n)u(n)$$

2. In addition to *M* first-order poles, *X*(z) also has an *N*-th order pole at $z=z_i$

$$X(z)=A_0+\sum_{m=1}^{M}\frac{A_m z}{z-z_m}+\sum_{j=1}^{N}\frac{B_j z}{(z-z_i)^j}\quad (|z|>\max\{z_1,\ldots z_M,z_i\})$$

$$\text{Let } X_1(z)=(z-z_i)^N\frac{X(z)}{z}$$

$$B_N=X_1(z)\big|_{z=z_i}\qquad B_j=\frac{1}{(N-j)!}\left[\frac{d^{N-j}}{dz^{N-j}}X_1(z)\right]_{z=z_i}\quad (j=1,\ldots,N-1)$$

$$\text{If it is a second-order pole } N=2:\quad X_1(z)=(z-z_i)^2\frac{X(z)}{z}$$

$$B_2=X_1(z)\big|_{z=z_i}\qquad B_1=\left[\frac{d}{dz}X_1(z)\right]_{z=z_i}$$

Inverse transform: $x(n)=A_0\delta(n)+(A_1z_1^n+\cdots+A_Mz_M^n+B_1z_i^n+B_2nz_i^{n-1})u(n)$

Example 8-3: Find the inverse transform of $X(z)=\dfrac{z^2}{z^2-4z+4}\ \left(|z|>2\right)$.

Solution: $x(n)$ is a **right-sided sequence.**

$$X(z)=\frac{z^2}{(z-2)^2}\qquad X_1(z)=\frac{X(z)}{z}=\frac{B_1}{z-2}+\frac{B_2}{(z-2)^2}$$

where

$$B_2=\left[(z-2)^2X_1(z)\right]_{z=2}=[z]_{z=2}=2,\quad B_1=\left\{\frac{d}{dz}\left[(z-2)^2X_1(z)\right]\right\}_{z=2}=1$$

then $X(z)=\dfrac{z}{z-2}+\dfrac{2z}{(z-2)^2}$

$$x(n)=2^nu(n)+n2^nu(n)=(n+1)2^nu(n)$$

Generally, *X*(z) can be written as:

$$X(z)=\frac{b_0+b_1z+\cdots b_{r-1}z^{r-1}+b_rz^r}{a_0+a_1z+\cdots+a_{k-1}z^{k-1}+a_kz^k}$$

When using the **partial fraction expansion** method to find the inverse transform, you need to master the inverse transforms of basic forms.

Note: The inverse transform is related to the ROC.

$$\frac{z}{z-z_m}(|z|>|z_m|)\Leftrightarrow z_m{}^n u(n)$$

$$\frac{z}{z-z_m}(|z|<|z_m|)\Leftrightarrow -z_m{}^n u(-n-1)$$

Discussion: Only **proper fractions** can be expanded by **partial fractions**, but the expanded form needs to be **multiplied by z** to have the basic form of the inverse z-transform mentioned above.

To perform partial fraction expansion on $\frac{X(z)}{z}$, $\frac{X(z)}{z}$ is required to be a proper fraction — which means ***k*≥*r* is needed to ensure *X*(z) converges at z=∞. The ROC of** the z-transform of **a causal sequence is |z|>R_x, and *k*≥*r*** is a **necessary and sufficient condition** for **convergence**.

Example 8-4: $X(z)=\dfrac{-5z}{3z^2-7z+2} \quad (\dfrac{1}{3}<|z|<2)$, find $x(n)$.

Two-Sided Sequence

Solution: $\dfrac{X(z)}{z}=\dfrac{-\frac{5}{3}}{z^2-\frac{7}{3}z+\frac{2}{3}}=\dfrac{-\frac{5}{3}}{\left(z-\frac{1}{3}\right)(z-2)}=\dfrac{1}{z-\frac{1}{3}}-\dfrac{1}{z-2}$

$$X(z)=\frac{z}{z-\frac{1}{3}}-\frac{z}{z-2}$$

Right-Sided Sequence

Left-Sided Sequence

$$x(n)=\left(\frac{1}{3}\right)^n u(n)+2^n u(-n-1)$$

The same z-transform expression with different ROCs correspond to different sequences.

$$X(z) = \frac{z}{z - \frac{1}{3}} - \frac{z}{z-2}$$

$$\text{if } |z| < \frac{1}{3} \qquad x(n) = -\left(\frac{1}{3}\right)^n u(-n-1) + 2^n u(-n-1)$$

$$\text{if } |z| > 2 \qquad x(n) = \left(\frac{1}{3}\right)^n u(n) - 2^n u(n)$$

Homework

1. Repeat all the examples in the lecture notes.
2. 7.17(e) and 7.24 (a) in the textbook.